Exponentials and Logarithms 3

Review

1. $y \propto x^2 \Rightarrow y = kx^2$ $0.5 = 4k \Rightarrow k = \frac{1}{8}$

$$\begin{aligned} y &= kx^2 \\ &= \frac{1}{8} \times 5^2 \\ &= \frac{25}{8} \end{aligned}$$

2.
$$\begin{aligned} \cos^2\theta + \cos\theta &= \sin^2\theta \\ \cos^2\theta + \cos\theta - 1 + \cos^2\theta &= 0 \\ 2\cos^2\theta + \cos\theta - 1 &= 0 \\ (2\cos\theta - 1)(\cos\theta + 1) &= 0 \\ \cos\theta &= \frac{1}{2} \text{ or } \cos\theta = -1 \end{aligned}$$

So $\theta = 60°, 180°, 300°$

3. d) $\frac{dy}{dx} = 0$ and $\frac{d^2y}{dx^2} < 0$

4.
$$\begin{aligned} \tan\theta &= \sqrt{3} \\ \theta &= \frac{1}{3}\pi, -\frac{2}{3}\pi \end{aligned}$$

5.
$$\int_1^2 (x^3 + 1)dx$$
$$= \left[\frac{x^4}{4} + x\right]_1^2 = \left(\frac{2^4}{4} + 2\right) - \left(\frac{1^4}{4} + 1\right)$$
$$= 6 - \frac{5}{4}$$
$$= \frac{19}{4}$$

Consolidation

1.
$$\begin{aligned} 2e^x - 5 - 3e^{-x} &= 0 \\ 2e^{2x} - 5e^x - 3 &= 0 \\ e^x &\in \left\{-\frac{1}{2}, 3\right\} \\ x &\in \left\{\ln\left(-\frac{1}{2}\right), \ln(3)\right\} \end{aligned}$$

But $\ln\left(-\frac{1}{2}\right)$ is undefined: $e^x > 0$.

So $x = \ln(3)$.

2.
$$\begin{aligned}\log_3(2x+p) &= \log_3(4x^2) - \log_3(3)\\ \log_3(2x+p) - \log_3\left(4\frac{x^2}{3}\right) &= 0\\ \log_3\left(\frac{6x+3p}{4x^2}\right) &= 0\\ \frac{6x+3p}{4x^2} &= 1\\ 4x^2 - 6x + 3p &= 0\end{aligned}$$

Since the equation has only one real root, the discriminant $b^2 - 4ac = 0$.

$$\begin{aligned}(-6)^2 - 4 \times 4 \times 3p = 0\\ 3p &= \frac{36}{16}\\ p &= \frac{12}{16}\\ &= \frac{3}{4}\end{aligned}$$

3. Each curve is the other curve reflected in the line $y = x$.
4. SKETCH SKETCH SKETCH SKETCH
5. a. If $2x - 1$ is a factor of $f(x) = 2x^3 - x^2 - 8x + 4$, $f\left(\frac{1}{2}\right) = 0$.

$$\begin{aligned}f\left(\frac{1}{2}\right) &= 2\left(\frac{1}{2}\right)^3 - \left(\frac{1}{2}\right)^2 - 8\left(\frac{1}{2}\right) + 4\\ &= \frac{1}{4} - \frac{1}{4} - 4 + 4\\ &= 0\end{aligned}$$

So therefore $2x - 1$ is a factor of $f(x)$.

i. Let $z = e^x$. Then

$$\begin{aligned}2z^3 - z^2 - 8z + 4 &= 0\\ (2z-1)(z^2-4) &= 0\\ (2z-1)(z-2)(z+2) &= 0\end{aligned}$$

So $e^x = \frac{1}{2}$ or $e^x = 2$.

$e^x \neq -2$ because $e^x > 0$.

Then

$$x = -\ln(2)$$
or
$$x = \ln(2)$$

Exam Questions

1. a. The student says $2(\log_3 x)^2 = 4\log_3 x$ when this is false, and the student says $\log_3 8 = 2$, when they most likely are thinking of $\log_2 8 = 3$. $\log_3 8 \neq 2$.

b. Let $z = \log_3 x$. Then

$$\begin{aligned} 2z^2 - 3z - 2 &= 0 \\ (2z+1)(z-2) &= 0 \end{aligned}$$

So $z = 2$ or $z = -\frac{1}{2}$. But $\log_3 x > 0$ so $\log_3 x = 2$. $x = 9$

2. a. Since $f(-1) = 0$, $(x+1)$ is a factor of $f(x)$. Then

$$\begin{aligned} x^3 + 4x^2 - 7x - 10 &= (x+1)(x^2 + 3x - 10) \\ &= (x+1)(x+5)(x-2) \end{aligned}$$

b. From part (a), we get that $x = -1$ or $x = -5$ or $x = 2$. If we let $x = e^y$ such that $x^3 + 4x^2 - 7x - 10 = e^{3y} + 4e^{2y} - 7e^y - 10$, then we get that $e^y \in \{-5, -1, 2\}$. But we know that $e^y > 0$, so $e^y \in \{2\}$.

Therefore, $e^y = 2$ and $y = \ln(2)$.

3. Let $a = e^x$ and $b = e^y$. Then

$$\begin{cases} a - 2b &= 3 \\ a^2 - 4b^2 = 33 \end{cases}$$

From equation 1, we get that

$$a = 3 + 2b$$

And therefore

$$\begin{aligned} (3+2b)^2 - 4b^2 &= 33 \\ 9 + 12b + 4b^2 - 4b^2 &= 33 \\ 12b &= 24 \\ b &= 2 \end{aligned}$$

Then since $a = 3 + 2b$, $a = 7$.

So

$$\begin{aligned} e^x &= 7 \\ x &= \ln(7) e^y = 2 \\ y &= \ln(2) \end{aligned}$$